

Machine Learning and Natural Language Processing Fall 2019

Homework 9

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Project Report

Word2Vec Demo

After successfully installing the `gensim` data the jupyter notebook throws a kernel error for the `wv = api.load('word2vec-google-news-300')` command. Running the command through a python interpreter also results in a memory error.

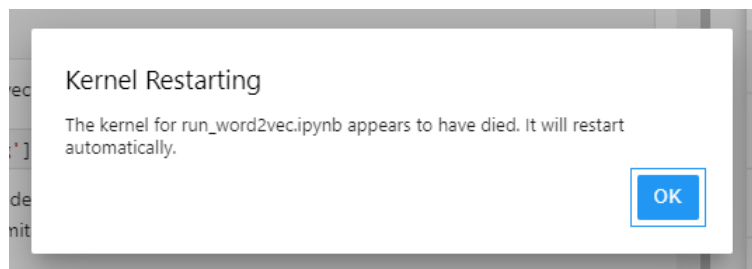


Figure 1: Jupyter Error

Training Your Own Model

Running the commands in the ‘**Training Your Own Model**’ section produces the following results on vocabulary retrieval:

```
hundreds
of
people
have
been
forced
to
their
homes
in
```

Evaluating

Below follows a sample of the evaluation section results (full results can be found in `run_word2vec_no_info.ipynb`)

```
[{'section': 'capital-common-countries',
 'correct': [],
 'incorrect': [('CANBERRA', 'AUSTRALIA', 'KABUL', 'AFGHANISTAN'),
 ('CANBERRA', 'AUSTRALIA', 'PARIS', 'FRANCE'),
 ('KABUL', 'AFGHANISTAN', 'PARIS', 'FRANCE'),
 ('KABUL', 'AFGHANISTAN', 'CANBERRA', 'AUSTRALIA'),
 ('PARIS', 'FRANCE', 'CANBERRA', 'AUSTRALIA'),
 ('PARIS', 'FRANCE', 'KABUL', 'AFGHANISTAN')],...]
```

Additionally, the model's performance when tested on the WS-353 set is the following:

```
((0.18441339927947975, 0.15837985274122884),
SpearmanrResult(correlation=0.1645129925406053, pvalue=0.20908536474072276),
83.0028328611898)
```

Examining the above performance results we conclude that the model has an almost 16% error rate (Pearson) and a p-value of 0.2 (Spearman).

Training Loss Computation

The training loss of our Word2Vec model in this instance is: 1380782.5

Benchmarks

The benchmark experiments for the trained model produce the following results:

```
Word2vec model #0: 'train_data': '25kB', 'compute_loss': True, 'sg': 0, 'hs': 0,
'train_time_mean': 1.8657992680867512, 'train_time_std': 0.11778208321728187
Word2vec model #1: 'train_data': '25kB', 'compute_loss': False, 'sg': 0, 'hs': 0,
'train_time_mean': 1.1016051769256592, 'train_time_std': 0.1540990541328008
Word2vec model #2: 'train_data': '25kB', 'compute_loss': True, 'sg': 0, 'hs': 1,
'train_time_mean': 2.397740364074707, 'train_time_std': 0.4653042813845671
Word2vec model #3: 'train_data': '25kB', 'compute_loss': False, 'sg': 0, 'hs': 1,
'train_time_mean': 2.7982991536458335, 'train_time_std': 0.08677591328510952
Word2vec model #4: 'train_data': '25kB', 'compute_loss': True, 'sg': 1, 'hs': 0,
'train_time_mean': 3.7357559204101562, 'train_time_std': 0.14055198227120438
Word2vec model #5: 'train_data': '25kB', 'compute_loss': False, 'sg': 1, 'hs': 0,
'train_time_mean': 3.763187885284424, 'train_time_std': 0.11876555861987233
Word2vec model #6: 'train_data': '25kB', 'compute_loss': True, 'sg': 1, 'hs': 1,
'train_time_mean': 6.7726108233133955, 'train_time_std': 1.1212763197079154
Word2vec model #7: 'train_data': '25kB', 'compute_loss': False, 'sg': 1, 'hs': 1,
'train_time_mean': 7.7274651527404785, 'train_time_std': 0.04394719618500559
Word2vec model #8: 'train_data': '1MB', 'compute_loss': True, 'sg': 0, 'hs': 0,
'train_time_mean': 4.8349818388621015, 'train_time_std': 0.21061255187130365
Word2vec model #9: 'train_data': '1MB', 'compute_loss': False, 'sg': 0, 'hs': 0,
'train_time_mean': 4.735631863276164, 'train_time_std': 0.4516237543679918
Word2vec model #10: 'train_data': '1MB', 'compute_loss': True, 'sg': 0, 'hs': 1,
'train_time_mean': 8.100348949432373, 'train_time_std': 1.1343235415319863
Word2vec model #11: 'train_data': '1MB', 'compute_loss': False, 'sg': 0, 'hs': 1,
```

```

'train_time_mean': 9.164145708084106, 'train_time_std': 0.24443668207075323
Word2vec model #12: 'train_data': '1MB', 'compute_loss': True, 'sg': 1, 'hs': 0,
'train_time_mean': 12.59971030553182, 'train_time_std': 1.577331655114504
Word2vec model #13: 'train_data': '1MB', 'compute_loss': False, 'sg': 1, 'hs': 0,
'train_time_mean': 12.500210603078207, 'train_time_std': 1.7662689273029815
Word2vec model #14: 'train_data': '1MB', 'compute_loss': True, 'sg': 1, 'hs': 1,
'train_time_mean': 26.202771186828613, 'train_time_std': 0.7012718184944973
Word2vec model #15: 'train_data': '1MB', 'compute_loss': False, 'sg': 1, 'hs': 1,
'train_time_mean': 23.495734691619873, 'train_time_std': 1.4624331722711774
Word2vec model #16: 'train_data': '10MB', 'compute_loss': True, 'sg': 0, 'hs': 0,
'train_time_mean': 60.69554082552592, 'train_time_std': 6.71928828338875
Word2vec model #17: 'train_data': '10MB', 'compute_loss': False, 'sg': 0, 'hs': 0,
'train_time_mean': 53.5072812239329, 'train_time_std': 1.2392807545587696
Word2vec model #18: 'train_data': '10MB', 'compute_loss': True, 'sg': 0, 'hs': 1,
'train_time_mean': 108.4642063776652, 'train_time_std': 5.774011006595503
Word2vec model #19: 'train_data': '10MB', 'compute_loss': False, 'sg': 0, 'hs': 1,
'train_time_mean': 115.19455862045288, 'train_time_std': 4.179655813764565
Word2vec model #20: 'train_data': '10MB', 'compute_loss': True, 'sg': 1, 'hs': 0,
'train_time_mean': 168.4000072479248, 'train_time_std': 7.695006067568017
Word2vec model #21: 'train_data': '10MB', 'compute_loss': False, 'sg': 1, 'hs': 0,
'train_time_mean': 164.73797472318014, 'train_time_std': 5.037241127942496
Word2vec model #22: 'train_data': '10MB', 'compute_loss': True, 'sg': 1, 'hs': 1,
'train_time_mean': 336.02881987889606, 'train_time_std': 1.16223784567851
Word2vec model #23: 'train_data': '10MB', 'compute_loss': False, 'sg': 1, 'hs': 1,
'train_time_mean': 222.32205971082053, 'train_time_std': 80.16010553951608
compute_loss hs sg train_data train_time_mean train_time_std
4 True 0 1 25kB 3.735756 0.140552
5 False 0 1 25kB 3.763188 0.118766
6 True 1 1 25kB 6.772611 1.121276
7 False 1 1 25kB 7.727465 0.043947
0 True 0 0 25kB 1.865799 0.117782
1 False 0 0 25kB 1.101605 0.154099
2 True 1 0 25kB 2.397740 0.465304
3 False 1 0 25kB 2.798299 0.086776
12 True 0 1 1MB 12.599710 1.577332
13 False 0 1 1MB 12.500211 1.766269
14 True 1 1 1MB 26.202771 0.701272
15 False 1 1 1MB 23.495735 1.462433
8 True 0 0 1MB 4.834982 0.210613
9 False 0 0 1MB 4.735632 0.451624
10 True 1 0 1MB 8.100349 1.134324
11 False 1 0 1MB 9.164146 0.244437
20 True 0 1 10MB 168.400007 7.695006
21 False 0 1 10MB 164.737975 5.037241
22 True 1 1 10MB 336.028820 1.162238
23 False 1 1 10MB 222.322060 80.160106
16 True 0 0 10MB 60.695541 6.719288
17 False 0 0 10MB 53.507281 1.239281
18 True 1 0 10MB 108.464206 5.774011
19 False 1 0 10MB 115.194559 4.179656

```

Adding Word2Vec “model to dict” method to production pipeline

```
2020-01-02 21:57:21,596 : INFO : precomputing L2-norms of word weight vectors
the [('an', 0.9999345541000366), ('and', 0.9999305009841919), ('in', 0.9999251365661621),
('which', 0.9999251365661621), ('of', 0.9999217391014099), ('with', 0.9999210238456726), ('on',
0.9999189972877502), ('first', 0.9999189972877502), ('for', 0.9999132752418518), ('its',
0.9999122023582458)]
to [('from', 0.9999515414237976), ('and', 0.9999467134475708), ('which', 0.9999457597732544),
('will', 0.9999457001686096), ('about', 0.9999446272850037), ('some', 0.9999443888664246),
('is', 0.9999427199363708), ('but', 0.9999426603317261), ('for', 0.9999414682388306), ('up',
0.999941349029541)]
of [('by', 0.9999473690986633), ('first', 0.9999438524246216), ('in', 0.9999423623085022),
('for', 0.9999391436576843), ('which', 0.9999366998672485), ('after', 0.9999357461929321),
('and', 0.9999350309371948), ('with', 0.9999344348907471), ('on', 0.999934196472168), ('says',
0.9999306201934814)]
```

The above results represent the ten most similar words for the first three terms in our dictionary.

Comparison with and without caching

Comparison without caching:

```
[('arafat', 0.9984195828437805), ('action', 0.9984142184257507), ('local', 0.9983998537063599),
('groups', 0.9983987808227539), ('anti', 0.9983932971954346), ('mr', 0.9983803033828735),
('received', 0.9983721375465393), ('since', 0.9983704090118408), ('called', 0.9983675479888916),
('interest', 0.998367190361023)]
[('up', 0.9998050332069397), ('have', 0.999800443649292), ('who', 0.9997922778129578), ('told',
0.9997885227203369), ('afghanistan', 0.9997878074645996), ('one', 0.999786376953125), ('over',
0.9997844099998474), ('two', 0.9997842311859131), ('and', 0.9997842311859131), ('if',
0.9997802972793579)]
[('his', 0.9999524354934692), ('with', 0.9999504089355469), ('up', 0.9999483227729797), ('from',
0.9999470710754395), ('as', 0.9999469518661499), ('which', 0.9999467134475708), ('into',
0.999946117401123), ('when', 0.9999450445175171), ('and', 0.9999444484710693), ('also',
0.9999440908432007)]
[('company', 0.9999383091926575), ('up', 0.9999348521232605), ('which', 0.9999284744262695),
('for', 0.9999249577522278), ('and', 0.999924898147583), ('out', 0.999923586845398), ('today',
0.9999215006828308), ('from', 0.9999208450317383), ('also', 0.999920129776001), ('after',
0.9999189376831055)]
0.05943036079406738
```

Comparison with caching:

```
[('arafat', 0.9984195828437805), ('action', 0.9984142184257507), ('local', 0.9983998537063599),
('groups', 0.9983987808227539), ('anti', 0.9983932971954346), ('mr', 0.9983803033828735),
('received', 0.9983721375465393), ('since', 0.9983704090118408), ('called', 0.9983675479888916),
('interest', 0.998367190361023)]
[('up', 0.9998050332069397), ('have', 0.999800443649292), ('who', 0.9997922778129578), ('told',
0.9997885227203369), ('afghanistan', 0.9997878074645996), ('one', 0.999786376953125), ('over',
0.9997844099998474), ('two', 0.9997842311859131), ('and', 0.9997842311859131), ('if',
0.9997802972793579)]
[('his', 0.9999524354934692), ('with', 0.9999504089355469), ('up', 0.9999483227729797), ('from',
0.9999470710754395), ('as', 0.9999469518661499), ('which', 0.9999467134475708), ('into',
0.999946117401123), ('when', 0.9999450445175171), ('and', 0.9999444484710693), ('also',
0.9999440908432007)]
```

